SYSTEMS ANALYSIS AND DESIGN / PHASE 2

Online Banking System

Context Model, Use Cases, and Activity Workflows

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The context and use case models are presented as lists of relationships. The activity diagrams are translated into step-by-step workflows.

Context model

The Online Banking System is the central system in the context model. It is connected to the following systems and services.

- Authentication system
- User account management system
- Financial transaction viewing and processing system
- Support system
- Shetab interbank network
- Civil registry system
- Users' web browsers
- Mobile operator systems
- Document upload system
- Sayad cheque system
- Electricity provider system
- Water provider system

Use case model

The user participates in all eight use cases. The external actors connected to each use case in the source diagram are listed below.

- Authenticate — Civil registry system.
- Recover password — User only in the use case model. The activity diagram also shows the mobile operator.
- Transfer funds — Shetab interbank network.
- Pay a bill — Shetab interbank network, water provider, and electricity provider.
- Register a Sayad cheque — Civil registry and Sayad cheque system.
- Apply for a loan — Document upload system.
- Purchase a mobile top-up — Shetab interbank network and mobile operators.
- Purchase a mobile data bundle — Shetab interbank network and mobile operators.

Activity diagrams

Authentication

1. Open the sign-in page and select an authentication level.

2. Level 1 — Enter a username and password. The online banking system checks the credentials. If valid, open the user page and grant access to corporate banking and the secure gateway. Otherwise, display an error and return.

3. Level 2 — Enter the card number, expiration date, CVV, and secondary card password. The Shetab network checks the details. If valid, open the user page and grant access to the customer club. Otherwise, display an error and return.

4. Level 3 — Enter a national ID and mobile number. The mobile operator and civil registry systems are involved in validation. If valid, open the user page and grant access to Sayad cheque management. Otherwise, display an error and return.

Password recovery

1. On the sign-in page, select Forgot Password and open the recovery page.

2. Enter the mobile number. If invalid, display an error and return. If valid, send a verification code through the mobile operator.

3. Enter the verification code. An invalid code returns the user to code entry.

4. Enter the new password twice. The banking system checks whether the password has already been used. If it has, return to password entry. Otherwise, update the password in the online banking system.

5. Display a success message and return to the sign-in page.

Fund transfer

1. Open the transfer page. The online banking system displays the transfer form.

2. Enter the destination card number or Sheba (Iranian IBAN) and the amount.

3. Request a one-time password through the Shetab network. The network also participates in verifying the transfer details.

4. If verification succeeds, debit the account and display a success message. If verification fails, display an error and return.

Bill payment

1. Open the Store, select Bills, and choose the bill type.

2. For a mobile bill, enter the bill ID, payment ID, and mobile number. The mobile operator system participates in processing the bill.

3. For other bills, enter the bill ID and payment ID.

4. In either branch, display a success message and return if payment succeeds. If it fails, display an error and return.

Sayad cheque registration

1. Open the Store, select Sayad Cheques, and choose New Operation.

2. Enter the Sayad cheque ID. If the Sayad system validates it, enable Continue. Otherwise, display an error asking the user to re-enter it.

3. Enter the amount and due date, then enable Continue.

4. Enter the beneficiary's national ID and mobile number. The civil registry and mobile operator systems participate in validation. If the details are invalid, display an error and ask the user to re-enter them.

5. Display the beneficiary's first and last name. Show the list of cheque purposes and let the user enter the purpose and description.

6. Show all the entered details. If the user declines confirmation, return to the previous step. If the user confirms, register the details and display the cheque in the Issued Cheques tab.

Loan application

1. Open the Store, select Loans, and choose a loan type.

2. Complete the application form. The document upload system handles supporting documents.

3. If the application data is invalid, display an error and ask the user to re-enter it. If valid, list the application under submitted loans.

4. If the bank approves the application, credit the loan amount to the user's balance. If it declines the application, display the reason.

Mobile top-up purchase

1. Open the Store, select Top-up, and enter the mobile number.

2. Display the amounts available from the mobile operator. Select an amount and enter the card details.

3. The Shetab network participates in card validation. If validation fails, display an error and return.

4. If validation succeeds, debit the balance, display a success message, and top up the user's SIM card.

Mobile data bundle purchase

1. Open the Store, select Mobile Data, and enter the mobile number.

2. Display the bundles available from the mobile operator. Select a bundle and enter the card details.

3. If validation fails, display an error and return.

4. If validation succeeds, debit the balance, display a success message, and apply the purchased bundle to the user's SIM card.